ZACHERY NOTT

Courtice, Ontario | znott@live.com | Portfolio: makerworld.com/@Azuya_1500

SUMMARY

Self-taught CAD designer with mechanical engineering coursework and hands-on experience in machine shops, fabrication, and product development. Designed and sold 3D printed products, led steering system design for a university concrete toboggan team, and worked as a municipal engineering inspector reviewing construction against drawings and standards.

TECHNICAL SKILLS

- CAD and Design: SolidWorks, Fusion 360, parametric and solid modelling, print-ready 3D modelling
- Fabrication: CNC, lathe, general machine shop, welding, 3D printing
- Engineering Practice: reading engineering drawings, inspection against standards, technical report writing
- Programming and Tools: Python, C, C++, SQL, HTML/CSS, Git/GitHub, Excel, Microsoft Office

PROJECTS AND SELF-EMPLOYMENT

Owner / Operator, Independent Small Engine and Equipment Repair January 2026 – Present
Clarington, ON

- Diagnose, troubleshoot, and repair small engines and outdoor equipment, including lawn mowers and trimmers
- Perform maintenance and repairs including carburetor cleaning, fuel system repairs, oil changes, spark plug replacement, and parts installation
- Source replacement parts and consumables from local and online suppliers to complete repairs efficiently
- Completed repairs for 5+ local customers, generating repeat business through quality workmanship

Owner and Designer, Independent Product Design Business January 2026 – Present

- Design original products using parametric CAD and manufacture prototypes and finished products through 3D printing
- Manage product development, online listings, customer orders, and fulfillment
- Published 13 original designs on online maker platforms and listed 8 products for sale
- Developed products including laboratory accessories, portable games, workshop tools, and hobby equipment

Steering Design Lead, GNCTR Concrete Toboggan Team 2021 – 2024
University of Ottawa

- Led design of the steering system for the team's concrete toboggan over 3 seasons
- Machined and welded components using CNC, lathe, and general shop equipment
- Designed and built the steering system after COVID-related team inactivity, improving vehicle performance and helping the team get back on its feet
- Produced design revisions based on testing results and fabrication constraints

WORK EXPERIENCE

Climbing Instructor (Seasonal), Junglesport Inc. April 2025 – Present
Ottawa, ON

- Deliver safety briefings and enforce safety protocols for all participants
- Coach technique, belaying, and knots with real-time feedback, building participant confidence

Paint Shop Labourer, General Motors Canada May 2023 – August 2023
Oshawa, ON

- Inspected vehicles for defects before painting and performed minor repairs
- Followed safety procedures and kept the work area clean and functional

Junior Inspector (Seasonal), WSP Global Inc. Summers: 2021, 2022, 2024
Whitby, ON

- Inspected contractor-installed underground services (sewer, water, plumbing) for compliance with approved drawings and standards
- Reviewed engineering drawings against the set standard
- Kept daily inspection logs and issued written findings to individuals and companies
- Drafted detailed reports with photographic documentation

Earlier Experience 2014 – 2020

- Lifeguard / swim instructor (Municipality of Clarington, Lifesaving Society, Legends Recreational Complex); snowboard instructor (Brimacombe); camp counsellor and lifeguard (CISS International Summer Camp); greenhouse labourer (Witzke's Greenhouses); paper carrier (Oshawa This Week), collecting 95% of voluntary payments

EDUCATION

University of Ottawa, Mechanical Engineering (B.A.Sc. program) 2020 – 2024
Degree paused; planning to complete remaining requirements.

- 1st Place, Design Day Competition (Medical VR Program)
- Engineering Students Society: 1st Year Representative; Manager of Sports

CERTIFICATIONS

Canadian Association of Snowboard Instructors (CASI) | Pleasure Craft Operator Card | National Lifesaving Certificate | Lifesaving and Swim Instructor Certificate | First Aid and CPR

VOLUNTEER

Golf Tournament Event Organizer, Reid James Foundation 2015 – Present

- Oversee registration and distribution of event items; guide patrons on the day

INTERESTS

Coding, design work, hockey, snowboarding and skiing, golf, reading, photography